ADITYA HARISH BOTHE

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SUMMARY

Manufacturing and industrial engineer with hands-on experience driving **process improvement**, **capacity analysis** and **cost reduction** across distribution and manufacturing operations. Six Sigma Green Belt certified, with hands-on experience in **root cause analysis**, **PFMEA** and control plans, equipment qualification, and building **Power BI** reporting that turns production data into decisions. Delivered an estimated **$3M** in savings and led a **$120K** capital project from concept through startup. Completing an MS in Industrial Engineering in December 2026.

EDUCATION

Master of Science in Industrial Engineering **Dec 2026**
Northeastern University, Boston 3.62 GPA

Bachelor of Engineering in Mechanical Engineering **Jul 2020**
Savitribai Phule Pune University, India 3.3 GPA

SKILLS

Process & Improvement: Lean, Six Sigma Green Belt, Kaizen, 5S, SMED, Value Stream Mapping, Standard Work, Time Studies
Quality & Validation: PFMEA, Control Plans, Root Cause Analysis, Corrective Action, IQ/OQ/PQ, SPC, Minitab, ISO 9001, 5-Why
Data & Systems: Power BI, SQL, Python, Advanced Excel, WMS (Manhattan Active, PKMS), Microsoft Office, Dashboard Automation
Design & Planning: SolidWorks, AutoCAD, Facility Layout, Material Flow, Capacity Planning, Capital Justification, Tooling Design

PROFESSIONAL EXPERIENCE

Supply Chain - Industrial Engineer Intern: Staples Inc, Framingham, MA **Jan 2026 – Sep 2026**

- Served as the sole on-site engineer supporting an automated fleet of more than 1,000 units across a multi-site distribution network
- Traced a fleet-wide stoppage to a single control dependency that nobody had treated as a point of failure, then engineered the fix
- Implemented **corrective and preventive action** on that failure, piloting then deploying a validated fix across every single unit
- Prevented recurring faults that carried an estimated **$100K per day** in direct operational exposure whenever they occurred
- Identified a facility capacity constraint through **capacity and constraint analysis**, delivering an estimated **$3M** in savings
- Developed **engineered labour standards** from detailed time study, recalibrating replenishment thresholds to cut trips **30%**
- Redesigned **facility layout** and material flow for an aisle expansion, unlocking space for more than 40 additional pallet positions
- Built **Power BI** dashboards giving leadership real-time operational visibility, migrating 650+ reports across four platforms
- Audited and corrected more than 400 reports to restore **data integrity**, establishing a reporting foundation the site could trust
- Traced order and material data flows between the **WMS** and material handling equipment to diagnose intermittent system faults
- Authored the detailed **runbook** and work instructions that let production teams troubleshoot independently on every shift
- Partnered across **Operations, Quality, Engineering and IT** teams to implement cross-functional process improvements on site
- Supported a new client site launch, personally onboarding more than 50,000 individual SKUs into the warehouse system
- Presented findings and improvement proposals to site leadership, refining each from their direct feedback on the floor
- Validated routing logic and cut times against operational requirements ahead of a major warehouse management system rollout
- Trained supervisors and associates on new processes and reporting tools following every system and process change

Manufacturing Engineer: AB Enterprises, India **Dec 2021 – Aug 2024**

- Owned process parameters across a 10-tank anodizing line, setting rectifier output at 13 to 17 V against calculated load surface area
- Led a **$120K** greenfield capital project from concept through to startup, procuring and commissioning 6+ machines in six months
- Raised daily output from 700 kg to a full ton per 8-hour shift through the greenfield plant build and facility layout design
- Designed **tooling and fixtures** in SolidWorks for multi-section part processing, raising overall batch capacity **40%** per load
- Verified coating thickness by DFT meter and bath chemistry by hydrometer, holding supplier calibration intervals on all instruments
- Oversaw production scheduling and raw material planning for an aluminium extrusion plant running 4 to 5 tons per day
- Cut unit cost **18%**, rework **10%** and unplanned downtime **25%** through process redesign and structured root cause analysis
- Resolved a customer quality escape on the customer's own site, tracing thickness and colour defects to one anodizing root cause
- Cross-trained more than 24 production associates across multiple distinct skill areas, building team flexibility through startup

PROJECTS

Process Risk and Quality System Analysis

- Documented a 10-tank process by **PFMEA**, scoring 8 failure modes and revising occurrence ratings against rejection data
- Built a linked 14-characteristic **control plan** defining measurement method, frequency and reaction plan per characteristic
- Assessed the operation against **ISO 9001** clauses 4 to 10, identifying conformance in calibration alongside open gaps

CERTIFICATIONS

Six Sigma Green Belt Specialization • Lean Concepts and Applications
Supply Chain Logistics • Operations Management: Quality and Supply Chain